

1 Problem Formulation

In the frequency domain, the Warburg impedance is given by

$$Z_W(\omega) = \frac{A_W}{\sqrt{j\omega}} \quad (1)$$

where A_W is the Warburg coefficient and ω is the angular frequency. Since Z_W has no finite-order rational form, it cannot be realized as a lumped circuit or driven directly in the time domain; we approximate it with an N -branch RC ladder network, then compare three time-domain approximators for integrating that network under an arbitrary current.

2 Methodology

2.1 RC Ladder Network

We approximate the Warburg element with N parallel $R_n C_n$ branches driven by a common current $i(t)$, with branch voltages summed:

$$Z_{RC}(\omega) \approx \sum_{n=1}^N \frac{R_n}{1 + j\omega R_n C_n}. \quad (2)$$

Rather than assigning the branch parameters $\{R_n, C_n\}$ from a closed-form scaling law, we solve for them directly: the $2N$ -parameter vector $\theta = (R_1, \dots, R_N, C_1, \dots, C_N)$ is obtained by nonlinear least squares so that the ladder's frequency response (2) matches the ideal Warburg impedance (1) across the frequency sweep $\{\omega_m\}_{m=1}^M$ used later for extraction (Section 2.3):

$$\theta^* = \arg \min_{\theta \geq 0} \sum_{m=1}^M |Z_{RC}(\omega_m; \theta) - Z_W(\omega_m)|^2. \quad (3)$$

The optimization is bounded ($R_n, C_n \geq 0$) and solved with a nonlinear least-squares routine (`scipy.optimize.curve_fit`), seeded from a log-spaced initial guess: time constants

$$\tau_n^{(0)} = \tau_{\min} \left(\frac{\tau_{\max}}{\tau_{\min}} \right)^{\frac{n-1}{N-1}}, \quad n = 1, \dots, N, \quad (4)$$

with initial values $R_n^{(0)} = A_W \sqrt{\tau_n^{(0)}}$, $C_n^{(0)} = \tau_n^{(0)} / R_n^{(0)}$ — the same $\sqrt{\tau}$ scaling that makes an unoptimized ladder trace an approximate 45° Nyquist line. Fitting (3) then refines this starting point into the branch values that minimize impedance error over the actual sweep band, rather than relying on the asymptotic argument that the scaling law only holds as $N \rightarrow \infty$ with densely spaced τ_n .

Model order selection ($N = 4$). To choose N , we refit (3) for $N \in \{2, \dots, 8\}$ and track the resulting RMS impedance error, $\text{RMSE}(N) = \sqrt{\frac{1}{M} \sum_m |Z_{RC}(\omega_m; \theta^*) - Z_W(\omega_m)|^2}$, alongside the

Bayesian Information Criterion evaluated on the stacked real/imaginary residual (Fig. 1). $\text{RMSE}(N)$ falls from $2.9 \text{ m}\Omega$ at $N = 2$ to $0.44 \text{ m}\Omega$ at $N = 4$, after which each additional branch pair buys a much smaller absolute reduction — only a further $0.43 \text{ m}\Omega$ combined across $N = 5$ through $N = 8$. We adopt $N = 4$ as a practical working order for this stage of the study: the fit error is already well below any expected experimental noise floor, the branch count stays small enough for straightforward time-domain implementation, and the optimizer still returns four distinct, monotonically ordered time constants spanning nearly four decades (2.8 ms to 4.8 s) that are physically interpretable as increasingly slow diffusion-like branches. A more rigorous order-selection criterion (e.g., BIC applied to noisy experimental data, where a finite-noise floor would penalize over-fitting) is left for future work.

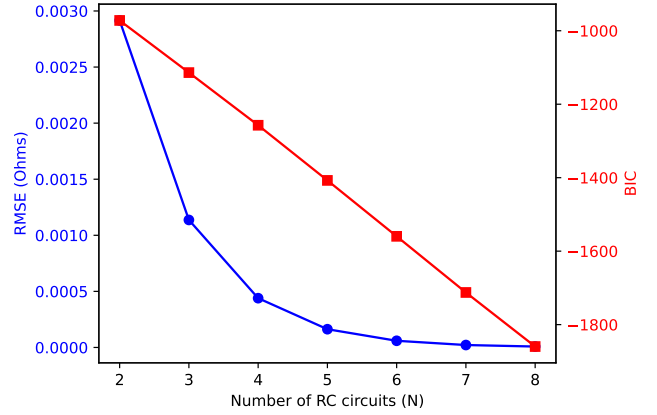


Figure 1: RMSE and BIC of the fitted RC ladder as a function of the number of branches N . Absolute RMSE improvement per added branch shrinks markedly beyond $N = 4$.

Each branch obeys a first-order linear ODE driven by the same $i(t)$:

$$\frac{dv_n}{dt} = \frac{i(t) - v_n/R_n}{C_n}, \quad n = 1, \dots, N, \quad (5)$$

and the total ladder (Warburg) voltage is

$$v_W(t) = \sum_{n=1}^N v_n(t). \quad (6)$$

Because branches are coupled only through the shared forcing $i(t)$, (5) is integrated independently and in parallel for all n , using the fitted θ^* from (3).

Frequency-dependent error validation. A single scalar RMSE can mask error that is concentrated in a narrow part of the spectrum, so before passing θ^* to the time-domain solvers introduced next, we also evaluate the

pointwise magnitude error

$$e(\omega_m) = \frac{|Z_W(\omega_m) - Z_{RC}(\omega_m; \theta^*)|}{|Z_W(\omega_m)|} \quad (7)$$

across the full swept band (Fig. 2). This confirms the $N = 4$ ladder holds a tight, well-behaved error tolerance over most of the frequency range of interest, rather than relying on an averaged metric alone that could obscure large local deviations near the edges of the sweep; error remains below 6% up to roughly 20 Hz before climbing toward the high-frequency edge of the band, where the excitation frequency exceeds the ladder's fastest fitted time constant.

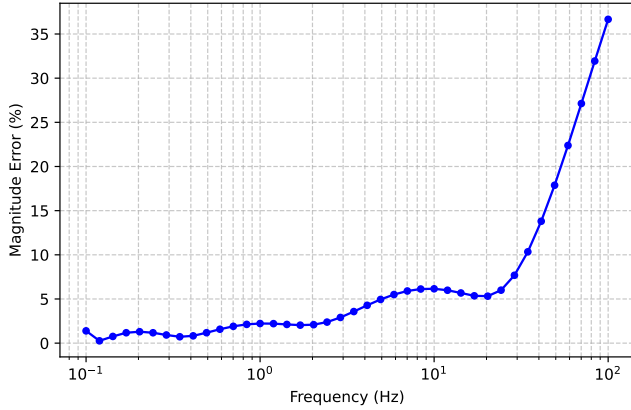


Figure 2: Magnitude error between the ideal Warburg impedance and the fitted $N = 4$ RC ladder across the swept frequency band.

2.2 Time-Domain Approximators

Given a sampled current $i[k] = i(kT_s)$ at step size T_s , we compare three ways of marching (5) forward — this is the actual point of comparison in this work, not the ladder topology itself.

2.2.1 Approach 1: Exact Zero-Order-Hold (ZOH)

Assuming $i(t)$ held constant at $i[k+1]$ over each interval, (5) has a closed-form per-branch solution:

$$v_n[k+1] = e^{-T_s/\tau_n} v_n[k] + R_n \left(1 - e^{-T_s/\tau_n}\right) i[k+1]. \quad (8)$$

This is exact for the assumed piecewise-constant input at any T_s/τ_n ; its only error source is how well that ZOH assumption represents the true continuous current between samples.

2.2.2 Approach 2: Forward Euler

$$v_n[k+1] = v_n[k] + T_s \frac{i[k] - v_n[k]/R_n}{C_n}. \quad (9)$$

First-order accurate ($O(T_s)$ local error); like (8), it uses only $i[k]$ over the whole step, with no mechanism for resolving how $i(t)$ varies within $[kT_s, (k+1)T_s]$.

2.2.3 Approach 3: RK4 with Interpolated Input

Classical fourth-order Runge–Kutta for the non-autonomous system (5) requires the forcing evaluated at four points within the step: t_k , $t_k + T_s/2$ (twice), and $t_k + T_s$. Since $i(t)$ is only known at $i[k]$ and $i[k+1]$, the midpoint value is obtained by linear interpolation,

$$i_{\text{mid}} = \frac{1}{2} (i[k] + i[k+1]), \quad (10)$$

giving stages

$$k_1 = f(v_n[k], i[k]), \quad (11)$$

$$k_2 = f(v_n[k] + \frac{T_s}{2} k_1, i_{\text{mid}}), \quad (12)$$

$$k_3 = f(v_n[k] + \frac{T_s}{2} k_2, i_{\text{mid}}), \quad (13)$$

$$k_4 = f(v_n[k] + T_s k_3, i[k+1]), \quad (14)$$

with $f(v, i) = (i - v/R_n)/C_n$, combined as

$$v_n[k+1] = v_n[k] + \frac{T_s}{6} (k_1 + 2k_2 + 2k_3 + k_4). \quad (15)$$

Unlike (8) and (9), this scheme does not treat $i(t)$ as frozen for the whole step: the interpolation in (10) tracks the continuously varying forcing within $[t_k, t_k + T_s]$ rather than approximating it by a single sample. This matters even for a clean, noise-free single-tone current $i(t) = \sin(\omega t)$ whenever T_s is not negligible relative to the drive period $1/f$: at the top of the swept band ($f = 100$ Hz, $T_s = 1$ ms, i.e. 10 samples per period), Approaches 1 and 2 accrue non-trivial error from holding $i[k]$ fixed across each step, while Approach 3's linear interpolation of the forcing keeps this discretization error an order of magnitude smaller. Because the current is otherwise a pure sinusoid, the resulting impedance-extraction error is attributable solely to the choice of time-domain integrator — a clean baseline discretization-error comparison, isolated from any input noise.

2.3 Frequency Response Extraction

For each drive frequency f , the ladder is excited with sampled current $i[k]$, the voltage $v_W[k]$ is produced by one of (8), (9), or (15), and the impedance is extracted after discarding the initial transient via a discrete Fourier transform at the bin nearest f :

$$\hat{Z}(f) = \frac{\text{FFT}\{v_W\}(f)}{\text{FFT}\{i\}(f)}. \quad (16)$$

Each approximator's accuracy is judged against the algebraic ladder impedance (2) evaluated at the fitted θ^* , which serves as ground truth independent of any time-domain discretization error.

3 Results and Discussion

Figure 3 overlays the three time-domain approximators against the algebraic ladder impedance (2) on a Nyquist plot. At this scale the three trajectories are visually indistinguishable from the ground truth across the swept band; the differences between integrators only become apparent when the per-frequency error is examined directly.

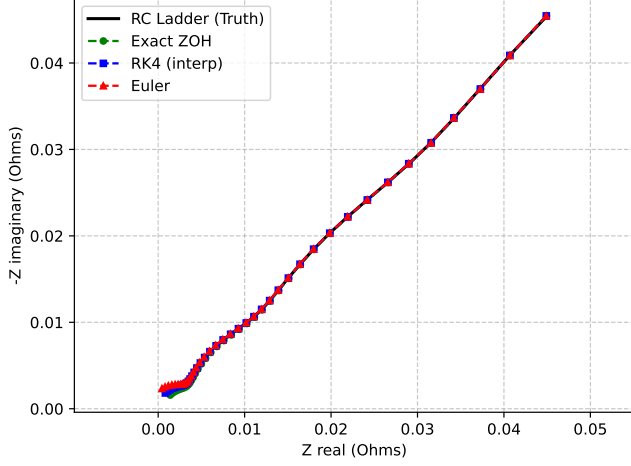


Figure 3: Nyquist plot comparing the three time-domain integrators against the RC-ladder impedance evaluated algebraically from (2).

Figure 4 plots this impedance-extraction error on a log-log scale for all three integrators. The interpolated RK4 scheme (Approach 3) maintains the lowest error across nearly the entire band, staying below roughly 1% through most of the sweep and reaching only about 3% at the top of the band (100 Hz). The exact ZOH and forward Euler schemes, which both treat the drive current as frozen at a single sample for the whole step, grow past 1% error within the first decade and exceed 20% by 100 Hz, where only ten samples span each drive period. This is consistent with the discretization argument in Section 2.3: for the fixed step size $T_s = 1$ ms used here, the dominant source of impedance-extraction error at high frequency is the choice of time-domain integrator rather than the RC-ladder approximation itself, which was already validated separately to sub-percent accuracy over most of the band (Fig. 2).

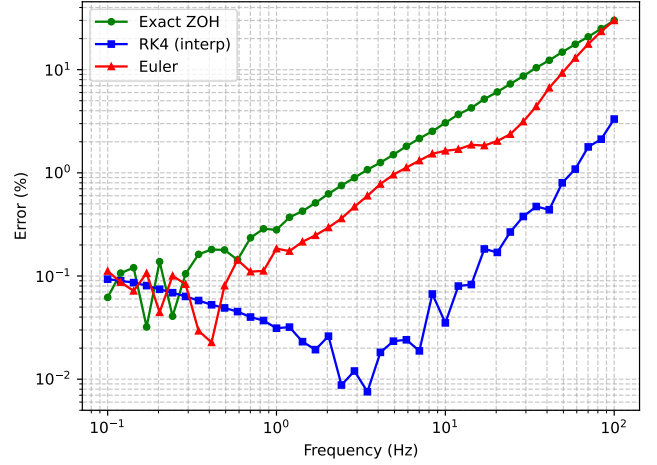


Figure 4: Impedance-extraction error (%) versus frequency for the three time-domain integrators, relative to the algebraic RC-ladder impedance.